Celestial Horizons

EU ROCKET LAB - BIP 2026

Electronic Architecture & Code Structure

Mission 1:
Hardware & Architecture

Designing a robust, compact, and optimized wiring schema for high-frequency data acquisition.

Mission Constraints

High Acceleration

The system must withstand extreme G-forces during the initial boost phase without losing sensor connectivity.

Severe Vibrations

Minimizing wiring via shared data buses (I2C/SPI) drastically reduces the risk of loose contacts during thrust.

Limited Volume

The electronics bay is highly constrained, requiring a compact vertical integration strategy without external heavy regulators.

The Processing Core

Heltec WiFi LoRa 32 V4

Acting as the central brain of our experimental rocket, this microcontroller handles all real-time processing and radio communication.

Power Management

- Powered by an 800mAh 3.7V LiPo Battery.
- Direct connection to the board's JST port.
- Built-in LDO regulates 3.7V to a clean 3.3V, eliminating the need for bulky external regulators.

Sensor Data Buses

I2C Protocol (Parallel)

Used by the **BME280** (environmental) and **MPU-6050** (inertial) sensors.

Both modules share the exact same 4 wires (VCC, GND, SDA, SCL). This daisy-chaining minimizes cable clutter and reduces structural vulnerabilities inside the bay.

SPI Protocol (High Speed)

Dedicated exclusively to the **Micro SD Reader**.

Unlike I2C, SPI offers massive bandwidth. This is a strict technical requirement to write MPU-6050 data at 200Hz during the boost phase without creating bottlenecks.

Strict Wiring Specification

Subsystem / Bus	Component Pin	Heltec V4 GPIO	Color Standard
I2C (Sensors)	VCC / GND	3.3V / GND	Red / Black
I2C (Sensors)	SDA / SCL	GPIO 17 / GPIO 18	Blue / Green
SPI (SD Card)	MISO / MOSI	GPIO 37 / GPIO 35	Yellow / Orange
SPI (SD Card)	SCK / CS	GPIO 36 / GPIO 5	Purple / White

Mission 1:
Embedded Software

Developing an autonomous Finite State Machine (FSM) to balance high-speed acquisition and radio transmission.

The Finite State Machine

1. IDLE

Pre-launch. Low power mode. MPU in detection mode. LoRa off.

2. BOOST

0-3s. Z-axis trigger. Max CPU to SPI writer. LoRa strictly disabled.

3. FLIGHT

Coast & Descent. BME280 active to detect apogee and track parachute trajectory.

4. RECOVERY

Landing confirmed (alt stable). Full LoRa activation for GPS and data dump.

The Critical Boost Phase

200

HERTZ (HZ)

Uninterrupted Data Logging

During State 2 (Propulsion), the ESP32 CPU is dedicated 100% to reading the MPU-6050 and writing to the SD card.

To achieve this 200 Hz sampling rate without processor lag, the LoRa radio chip remains completely disabled until the rocket safely lands.

Data Transmission Strategy

1. In-Flight

Minimalist approach. Simple beacons every 10s with altitude and battery. High-freq data remains stored locally.

2. Touchdown

Immediate emergency transmission containing GPS coordinates and max altitude to locate the capsule.

3. Data Dump

Differed transmission. The ESP32 slowly reads the SD CSV file and transmits the complete 3D flight history.

Energy Management

- **Pre-Launch Conservation:** The ESP32 is kept in "Light Sleep" mode on the pad, saving the 800mAh battery for the actual flight.
- **Dynamic Sensor Activation:** Sensors are polled adaptively. High frequency is only engaged during Boost, dropping to lower polling rates during descent.
- **Local Storage Priority:** By recording raw data to the SD card instead of transmitting it via LoRa in real-time, we drastically cut down radio-induced power consumption.
- **Post-Landing Radio:** The energy-intensive LoRa module is fully awakened only when acceleration hits zero for 5 seconds, ensuring enough power remains for the recovery beacon.

Questions?

Thank you for following the Celestial Horizons presentation.

Image Sources

https://hackster.imgix.net/uploads/attachments/1704385/image_iGT0JMQMt7.png?auto=compress%2Cformat&w=1200

Source: www.hackster.io

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https://www.dailyclimate.org/media-library/a-rocket-launch-with-a-blue-sky-and-clouds-in-the-background.jpg?id=61508702&width=2000&height=1500&coordinates=35%2C0%2C35%2C0

Source: www.dailyclimate.org

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